REVIEW

Update in clinical and mouse microbiota research in allogeneic haematopoietic cell transplantation

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Purpose of review

The intestinal microbiota plays a critical role in intestinal homeostasis and immune regulation and has been recognized as a predictor of clinical outcome in patients undergoing allogeneic haematopoietic cell transplantation (allo-HCT) and specifically a determinant of the severity of graft-versus-host disease (GVHD) in mouse models. As GVHD is the most important cause of nonrelapse mortality (NRM) after allo-HCT, understanding the mechanisms by which modifying the microbiota may prevent or decrease the severity of GVHD would represent an important advance.

Recent findings

Microbiota injury was observed globally and higher diversity at peri-engraftment was associated with lower mortality. Lactose is a dietary factor that promotes post-allo-HCT Enterococcus expansion, which is itself associated with mortality from GVHD in patients and exacerbates GVHD in mice. Bacterial and fungal bloodstream infections are preceded by intestinal colonization with a corresponding organism, supporting the gut as a source for many bloodstream infections. Metabolomic profiling studies showed that GVHD is associated with changes in faecal and plasma microbiota-derived molecules.

Summary

In this review, we highlight some of the most recent and important findings in clinical and mouse microbiota research, as it relates to allo-HCT. Many of these are already being translated into clinical trials that have the potential to change future practice in the care of patients.

Keywords

allogeneic haematopoietic cell transplantation, clinical outcome, microbiota, preclinical studies

INTRODUCTION

Allogeneic haematopoietic cell transplantation (allo-HCT) is a curative treatment for haematologic malignancies in which patients receive a conditioning regimen followed by infusion of haematopoietic cells from a donor. Overall mortality after allo-HCT is in the order of 50% and is largely attributable to relapse of the underlying disease, graft-versus-host disease (GVHD) and infections. Allogeneic donor T and natural killer (NK) cells can mediate protective graft-versus-tumour (GVT) activity, but alloreactive donor T cells can also recognize normal host tissues resulting in GVHD. GVHD is initiated by conditioning-regimen-induced intestinal inflammation and gut-barrier defects that allow transfer of bacterial molecules across the damaged gut epithelium [1]. Although the gut microbiota is well recognized as the source of these bacterial products, a hypothesis that underpins much work in the field is that the microbiota also modulates additional steps in the pathophysiology of GVHD, perhaps including priming and differentiation of donor T cells or subsequent innate immune inflammatory mediators. Here, we focus on the most recent clinical and mouse studies in the context of microbiome and allo-HCT. Several authors have written excellent comprehensive reviews on the subject [2–8].

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KEY POINTS

- Intestinal microbiota disruption occurs in patients undergoing allo-HCT across transplant centres and geographic locations; higher diversity during the periengraftment period is predictive of better outcome.
- Enterococcus domination is common after allo-HCT, is promoted by dietary lactose and accelerates GVHD in mice
- Bacterial and fungal bloodstream infections in allo-HCT recipients are preceded by intestinal colonization with corresponding organisms, supporting the gut as a source for disseminated infection.
- Microbiota-derived metabolites may mitigate or promote GVHD.
- MHC class II expression and antigen presentation by intestinal epithelial cells is influenced by peritransplant intestinal microbiota; this is responsible for the initiation of lethal GVHD in mice.

MAIN

Microbiota diversity and graft-versus-host disease related mortality

In patients undergoing allo-HCT several factors such as diet, antibiotics, conditioning regimen, mucositis, GVHD and infections are hypothesized to influence microbiota composition [6]. One key measure of microbiota disruption, α-diversity, reflects the number of unique bacterial taxa in a faecal sample as well as their relative abundances. The first multicentre study profiled 8767 stool samples from 1362 adult allo-HCT recipients by means of 16S rRNA gene sequencing and correlated features of the intestinal microbiota with clinical outcomes retrospectively [9▪]. The key observations reported by this international collaborative study (coordinated at our centre) were that microbiota injury, which was characterized by loss of diversity and domination by single taxa (most commonly the genus Enterococcus) occurred across four transplant centres on three continents. Above-median diversity at periengraftment (day 7–21) was associated with lower risk of death, mostly related to GVHD in a subgroup analysis. Finally, baseline faecal samples (day -30 to -6) already showed patterns of microbiota injury compared with healthy volunteers and below-median diversity was predictive of poor survival, even when this was only present prior to transplant. These observations seemed to occur independently of differences in clinical practices between transplant centres and of geographic variations in the microbiota composition. These results were concordant with other recent independent analysis [10].

We also derived a risk-score predictive of mortality based on the abundances of specific bacterial taxa [9▪], suggesting that in addition to diversity, specific bacterial abundances may be predictive of clinical outcome after allo-HCT. A separate single-centre study of 141 allograft recipients derived a risk score composed of day 15 faecal microbiota diversity and abundances of four bacterial families to predict grade II-IV aGVHD [11]. In another study, the same group also used linear regression to derive a gut microbiome score that was predictive of grade II-IV aGVHD in both a discovery and a validation set [12▪]. The strongest predictor within this score, which was based on 20 gut microbiota features measured at day 15, was the abundance of Lachnospiraceae. This observation is strikingly consistent with the observation that Lachnospiraceae are associated with less GVHD [13], and with our previous report that Blautia, a prominent genus within family Lachnospiraceae was predictive of less GVHD-related mortality [14]. Moreover, as an indirect measure of gut Lachnospiraceae abundance, urine concentrations of a metabolite produced by this group, 3-indoxyl-sulfate, were previously associated with less transplant-related mortality [15].

These risk scores are best considered as proof of principle that faecal microbiota composition carries some useful predictive information about patient outcome, and specific features that make key contributions to the risk scores can inform hypotheses to test in preclinical models.

Enterococcus domination

A high incidence of faecal community domination by Enterococcus in patients undergoing allo-HCT was previously observed in several studies [9▪,16–18]. Profiling of faecal samples of 1325 adult allo-HCT patients showed intestinal Enterococcus dominated of up to 65% allo-HCT patients early after transplant and was associated with a higher risk of lethal aGVHD [19▪]. Higher relative abundance of Enterococcus in the month following allo-HCT was independently reported to be associated with reduced survival in a cohort of 38 adult patients [20]. Although in patients, antibiotics are understood to be a major driver of microbiota injuries, including and Enterococcal expansion [17,18,21], we also observed an expansion of Enterococcus approximately 1 week after HCT in mouse models of GVHD without any antibiotic treatment [19▪]. Not only did Enterococci expand, but treatment of germ-free mice with Enterococcus caused an aggravation of GVHD and accelerated death. Using metagenomic shotgun sequencing, an enrichment of genes encoding enzymes involved in lactose utilization in the Enterococcus-dominated, posttransplant microbiota, which called attention to the possibility that lactose could be a sugar source for these bacteria. A lactose-free diet attenuated the Enterococcus bloom and GVHD severity in mice. This finding is an example of the potential for microbe-targeted dietary interventions to prevent aGVHD.

Interestingly, a recent study revealed, by using whole-genome sequencing of Enterococcus isolates from serial faecal samples, that Enterococcus domination is not a monotonous state but rather one of complex clonal dynamics [22▪]. In line with this, Zlitni *et al.* [23▪] observed that an apparently stable microbiota composition as assessed by 16S rRNA amplicon sequencing in an allo-HCT recipient actually harboured strain-level differences revealed by an innovative computational analysis of metagenomic DNA and RNA sequences. Importantly, both of these studies observed changes in antibiotic-resistance genes during the time courses that were examined.

Another important point to consider regarding Enterococci is that not all members of this genus are necessarily detrimental. Rashidi *et al.* [24,25] have published a series of studies focusing on *Enterococcus gallinarum* and *Enterococcus casseliflavus* in which surprising associations with improved survival were observed for these species. In concordance with these findings, in our risk-score for all-cause mortality [9▪], while Enterococcus as a genus was associated with a higher risk of mortality; *E. gallinarum* was actually associated with longer survival.

A lantibiotic protein produced by a strain of *Blautia producta* has recently been demonstrated to attenuate the growth of vancomycin-resistant Enterococci (VRE) [26▪]. Interestingly, this strain belongs to the genus that Jenq *et al.* [14] previously associated with lower GVHD-related mortality and to the family Lachnospiraceae discussed above. It is unknown whether Blautia has immunosuppressive properties and directly suppresses GVHD, whether Blautia works to suppress GVHD by attenuating Enterococcus or whether Blautia exerts no effect but is simply a reproducibly observed biomarker of microbiota and general gastrointestinal health in allo-HCT patients.

Microbiota and infections

Allo-HCT patients are at a high risk for bloodstream infections (BSIs), especially during the posttransplant neutropenic period, when mucosal injury is at its most severe and domination of the intestinal microbiota with particular pathogenic genera occurs [17]. Exploring the longstanding hypothesis that intestinal bacteria are the source of many BSI following stem cell transplantation, Tamburini *et al.* [27] identified that patients with culture-confirmed *Escherichia coli* and *Klebsiella pneumoniae* BSI had concomitant gut colonization with matching strains of these bacteria. In line with this study, a prospective study of nine paediatric allo-HCT patients matched BSI isolates to faecal samples [28▪]. Another analysis of a large cohort of 708 allo-HCT patients with 4768 faecal samples found that gram-negative intestinal domination was associated with subsequent BSI, both peaking 4–5 days posttransplantation. Together, these are consistent with the notion of a causal pathway of gram-negative BSI through mucosal injury and translocation [29▪].

Of note, exposure to fluoroquinolone prophylaxis has been associated with a lower risk of gram-negative BSI, but higher relative proportion of intestinal domination and BSI with *E. coli* and specifically increased fluoroquinolone resistance [29▪]. This suggests a role for antibiotic prophylaxis regimens in shaping microbiota composition [30] including intestinal domination and subsequent infection risk. In contrast, however, no reduction in microbiota diversity was observed in a comparison of microbiota composition of faecal samples collected during no-antibiotic versus levofloxacin-exposed intervals [31]. In recently updated recommendations for antimicrobial prophylaxis [32], fluroquinolone prophylaxis was recommended at least for recipients of myeloablative conditioning. The benefits and risks of such prescribing practices are a matter of ongoing debate [33,34], as antibiotic exposure has been associated with several adverse outcomes [35], including GVHD and GVHD-related mortality [36–41], infections [17,29▪,42,43], antibiotic resistance [44,45]. Slavin *et al.* [33] have pointed out that the evidence base supporting antibacterial prophylaxis was accumulated prior to the widespread adoption of now-standard vigilant recognition of sepsis and immediate administration of empiric antibiotics, perhaps decreasing the rationale for prophylaxis in favour of other goals such as preservation of microbiome health and antimicrobial stewardship.

The relationship between the fungi that reside in the gut – that is, the mycobiota – and risk of invasive fungal infection is less-well studied. In a proof-of-concept study, Zhai *et al.* [46▪] showed that candidemia was preceded by rapid expansion of pathogenic *Candida spp.* in the gut, in striking similarity to the observations that have been made in bacteria. Furthermore, intestinal *Candida spp.*

domination was correlated with loss of bacterial diversity and exposure to activity against anaerobic bacteria. *Blautia producta* and *Bacteroides thetaiotaomicron* were used as markers for a heathy gut bacteria community and correlated inversely with fungal relative abundance, alluding to close interactions between bacterial microbiota and mycobiota. These observations reinforce again that a healthy microbiome may protect against infectious complications after allo-HCT such as enteric BSI, *Clostridioides difficile* [47] and respiratory viral infections [42,43].

However, developing microbiota-based assays to predict and prevent BSI in allo-HCT recipients has several challenges: frequent faecal sampling would be necessary to sensitively predict BSI in allo-HCT patients because of a high microbiota volatility in the immediate period after transplant [21], cut-off thresholds for specific strains may vary and additional samples from other body sites might be necessary. A novel approach to predict infections may be offered by an assay of microbial cell-free DNA that is approved for clinical use in some locations in the U.S. [48]. Use of this assay in adult and paediatric HCT recipients [49–52] suggests that it can be implemented in clinical practice and may offer a strategy to reduce use of broad-spectrum antibiotics and to spare the intestinal microbiome. However, larger studies to evaluate this approach are needed.

Microbial metabolites

Microbial metabolites – which are intermediate or end products of microbial metabolism derived from diet, host molecules or directly from bacteria – influence host intestinal homeostasis, energy metabolism and immunity [53]. These molecules can be profiled via mass spectrometry-based and chromatography-based techniques and their importance to microbiota-host interactions in general has been described in clinical and preclinical studies. A recent high-throughput metabolomic analysis of plasma metabolites in patients at the onset of aGVHD in comparison with transplant recipients who did not get GVHD ($N = 38$ cases and $N = 62$ controls were compared) showed significant variation in microbial metabolic pathways, including those related to amino acids (tryptophan and arginine), lipids (lysoplasmalogens, plasmalogens, and phospholipids) and complex lipid products (medium and long-chain fatty acid, polyunsaturated fatty acid and bile acids) [54▪].

In GVHD, the short-chain fatty acid (SCFA) butyrate is the most studied of the microbial metabolites. In mice, butyrate-producing bacteria have been shown to reduce GVHD [55,56], and in humans, a higher abundance of butyrate-producing bacteria have been separately correlated with lack of exposure to antibiotics with less GVHD-related mortality [14] and fewer respiratory viral infection after allo-HCT [42,43].

A recent retrospective study with prospective weekly stool collection from 316 patients compared patients sampled at aGVHD onset ($N = 35$) with non-GVHD patients ($N = 35$) and showed that faecal butyrate concentration was reduced at aGVHD onset in any stage of gastrointestinal aGVHD, while faecal concentrations of the SCFAs propionate and acetate were only reduced in severe gastrointestinal aGVHD, suggesting faecal butyrate may be a potential aGVHD marker for diagnosis and faecal propionate and acetate concentration for severity at the time of GVHD onset [57▪]. This is in agreement with a previous study in paediatric allo-HCT patients, in which the authors reported lower faecal butyrate concentrations on day 14 were associated with aGVHD when they compared children who do developed GVHD and children who do not develop GVHD [41]. Butyrate concentration in plasma was also measured in a clinical trial of human chorionic gonadotropin for steroid-refractory GVHD. Responders had higher plasma butyrate concentrations at study day 28 and 56 compared with baseline, while no change in butyrate concentration over time was observed in nonresponders [58▪]. In a case–control study that measured plasma butyrate concentrations peri-day 100 after allo-HCT, lower plasma butyrate and propionate concentration were associated with the subsequent development of cGVHD [59▪]. Interestingly, in another study, increased relative abundance and presence of at least one bacterium predicted to produce butyrate within 21 days after the onset of aGVHD was associated with developing steroid-refractory GVHD and cGVHD [60▪]. In experiments using human intestinal organoids, the authors observed that butyrate inhibits colonic stem cells from forming an intact epithelial layer [60▪]. One way to synthesize these disparate observations may be a model in which butyrate is protective before the onset of aGVHD perhaps at the level of modulating T cell biology, but after the onset of aGVHD, it attenuates intestinal recovery by inhibiting colonic stem cell proliferation and healing. Another way may be dose dependency, as high amounts of short-chain fatty acids fail to ameliorate GVHD in the mouse system [56].

Not only does the gut microbiome incur an injury during transplantation, but the oral communities are also disrupted. One study identified oral microbiota injury patterns and, recently, the decreased production of polyamines by the microbiome was associated with higher rates of oral mucositis in a cohort of HCT recipients [61].

Table 1. Ongoing clinical trials

Study title	Intervention	Primary endpoint	NCT number
GVHD treatment			
Fecal Microbiota Transplantation for Steroid Resistant/Dependent Acute GI GVHD (FEMITGIGVHD)	FMT	Response rate of GI aGVHD	NCT03812705
Fecal Microbiota Transplantation (FMT) in Recipients After Allogeneic Hematopoietic Cell Transplantation (HCT)	FMT	Gut microbiome diversity	NCT03720392
Fecal Microbiota Transplantation in aGvHD After ASCT	FMT	GI aGvHD remission	NCT03819803
Treatment of Steroid Refractory Gastro-intestinal Acute GVHD afteR AllogeneiC HSCT With fEcal Microbiota tranSfer (HERACLES)	FMT	GI and overall aGVHD response	NCT03359980
FMT In High-Risk Acute GVHD After ALLO HCT	FMT	Proportion of patients who are able to swallow the specified number of capsules	NCT04139577
Fecal Microbiota Transplantation With Ruxolitinib and Steroids as an Upfront Treatment of Severe Acute Intestinal GVHD (JAK-FMT)	FMT Ruxolitinib	Overall survival	NCT04269850
Fecal Microbiota Transplantation for Steroid Resistant and Steroid Dependent Gut Acute Graft Versus Host Disease	FMT	Serious adverse events	NCT03214289
Fecal Microbiota Transplantation For The Treatment Of Gastro-Intestinal Acute GVHD	FMT	Toxicity Efficacy	NCT04059757
GVHD prevention			
Gut decontamination			
Gut decontamination in pediatric allogeneic hematopoietic	Vancomycin-polymyxin B	Faecal microbiota diversity	NCT02641236
Probiotic			
Lactobacillus plantarum in preventing acute graft versus host disease in children undergoing donor stem cell transplant	Lactobacillus plantarum strain 299 and 299v	Incidence of stage 1–4 GI aGVHD	NCT03057054
CBM588 in Improving Clinical Outcomes in Participants Who Have Undergone Donor Hematopoietic Stem Cell Transplant	*Clostridium butyricum* CBM 588 Probiotic Strain	Incidence of adverse events Feasibility	NCT03922035
Prebiotic			
Dietary Manipulation of the Microbiome-metabolomic Axis for Mitigating GVHD in Allo HCT Patients	Potato-based starch	Incidence of grade II-IV aGVHD	NCT02763033
Human Lysozyme Goat Milk for the Prevention of Graft Versus Host Disease in Patients With Blood Cancer Undergoing a Donor Stem Cell Transplant	Human lysozyme goat milk	Patients' ability to drink the specified daily amount human lysozyme goat milk Unacceptable toxicity Adverse events Volume of human lysozyme goat milk consumed	NCT04177004
The Use of A Prebiotic to Promote a Healthy Gut Microbiome in Pediatric Stem Cell Transplant Recipients	Inulin	Change in α and β−bacterial diversity measures in stool Change in faecal short chain fatty acid concentration	NCT04111471
Oral Supplementation of 2′-Fucosyllactose in Allogeneic Bone Marrow Transplant Recipients	2′-fucosyllactose	Number of BSI Number of patients able to take 2′-fucosyllactose	NCT04263597
Infection prophylaxis			
Rifaximin for Infection Prophylaxis in Hematopoietic Stem Cell Transplantation	Rifaximin	Faecal microbiota diversity	NCT03529825

Table 1 (*Continued*)

Study title	Intervention	Primary endpoint	NCT number
Choosing the Best Antibiotic to Protect Friendly Gut Bacteria During the Course of Stem Cell Transplant	Piperacillin-tazobactam versus cefepime	Change in Clostridiales abundance	NCT03078010
Optimization of Antibiotic Treatment in Hematopoietic Stem Cell Receptors (Optimbioma)	Optimization/ antibiotic strategy	Impact on microbiota	NCT03727113
Autologous Fecal Microbiota Transplantation (Auto-FMT) for Prophylaxis of Clostridium Difficile Infection in Recipients of Allogeneic Hematopoietic Stem Cell Transplantation	FMT	*Clostridioides difficile* infection	NCT02269150

Collectively, these findings suggest that microbiota-derived metabolites may play a significant role the pathogenesis of GVHD, though more mechanistical studies need to be conducted.

Microbiota and graft-versus-host-disease pathogenesis

Mechanisms by which gut microbiota may influence GVHD are actively being investigated. Koyama *et al.* [62] recently demonstrated in mouse studies that antigen presentation by intestinal epithelial cells to donor T cells plays a critical role in GVHD pathogenesis. The ability of intestinal epithelial cells to execute this was dependent on the MyD88/TRIF signalling pathway, indicating that it is induced by microbial signals [63]. A study from the same group evaluated the impact of the peri-transplant gut microbiome on GVHD, making use of the endogenously dysbiotic microbiota of mice deficient in IL17RA [64], which has been shown to increase GVHD [65]. Cohousing of wild-type mice with these mice was insufficient to increase GVHD in mice; however, cohousing posttransplant increased GVHD and continuously cohousing had an even bigger effect on GVHD, providing yet another example of the ability of microbiome to modulate GVHD severity.

Clinical strategies targeting the microbiome

The scientific and clinical challenges of engineering and implementing microbiota-based strategies into clinical practice are an active area of investigation in the field and are reviewed in detail [66]. Several clinical studies are ongoing (Table 1) in the areas of GVHD treatment and prevention and making use of diverse strategies, including gut decontamination, prebiotics, probiotics and optimization of antibiotic regimens.

Conclusion and future directions

Investigators continue to extend our understanding of the relationship between the intestinal microbiota and clinical outcomes after allo-HCT, and microbiota-associated GVHD pathogenesis including microbiota-based strategies. Numerous experiments have demonstrated manipulation of the gut microbiota can ameliorate or exacerbate GVHD severity in mice, indicating a causal relationship between microbes and alloreactivity. However, still unexplained is precisely how the microbiome modulates alloreactivity in mice and whether this occurs in humans. The role of intestinal microbiota not only in GVHD but also in the context of GVT [67] as well as the role of microbial communities at other body sites in allo-HCT related outcomes [61] offer important avenues for future study.

Because of the complexity of influencing factors on intestinal microbiota and clinical outcomes in patients undergoing allo-HCT randomized, prospective multicentre trials are essential and clinical trials will help to determine the role of microbiota-based strategies in clinical practice. Advances in diagnostic techniques and in the rational selection or design of probiotics may allow earlier and safer interventions. In addition, more mouse studies using evolving techniques are needed to gain more insights in molecular mechanisms.

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Conflicts of interest

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